P-D-1801
February 9, 1973

FEDERAL SPECIFICATION

DISHWASHING COMPOUND, LIQUID SYSTEM

(FOR COMMERCIAL DISHWASHING MACHINES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers free-flowing, solid and liquid form dishwashing compounds for use only with liquid-type detergent dispensers (see 6.1).

1.2 Classification. The dishwashing compounds shall be of the following types, classes, styles, and size containers, as specified (see 6.2).

Type I - Two-part liquid system dishwashing compound.
Part A - Dishwashing compound, 19 Gallon drum.
Part B - Water conditioning compound, 20 Gallon drum.

Type II - Single system dishwashing compound.
Class 1 - Liquid form.
Style A - With chlorine.
Style B - Without chlorine.
Style C - Non-phosphate.
6 gallon pail
15 gallon drum
30 gallon drum
Class 2 - Powdered form (granular) 50 pound pail.
Style A - With chlorine.
Style B - Without chlorine.
Style D - Without chlorine (low phosphate).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 7930

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Federal Specifications:

QQ-A-250/2	-	Aluminum Alloy 3003, Plate and Sheet
QQ-S-766	-	Steel Plates, Sheets and Strip, Corrosion-Resisting
RR-S-366	-	Sieve, Test
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood Cleated-Plywood.
PPP-B-636	-	Boxes, Shipping, Fiberboard.

Federal Standards:

FED-STD-123	-	Marking for Domestic Shipment (Civil Agencies)
FED-STD-141	-	Paint, Varnish, Lacquer and Related Materials, Methods of Inspection Sampling and Testing
FED-STD-536	-	Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaporproof Or Waterproof, Flexible.

Military Standards:

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-147	-	Palletized Unit Loads For 40" x 48" Pallets.

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(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

21 CFR 191 - Federal Hazardous Substances Act.

(The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to Association of Official Analytical Chemists, P.O. Box 540, Benjamin Franklin Station, Washington, D.C. 20044.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 So. Riverside Plaza, Chicago, Illinois 60606.)

American Society For Testing and Materials

ASTM E70-68 - pH of Aqueous Solutions with the Glass Electrode-Test For

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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3. REQUIREMENTS

3.1 Material. The type I and type II, class 1 liquid dishwashing compound and type I part B water conditioner, shall be uniform free-flowing liquid. The liquids shall be free of sediment, contain alkaline salts, phosphates, chlorine releasing compounds, and a liquid vehicle. The type II, class 2, powdered dishwashing compounds intended for use in liquid type dispensers shall be uniform, free flowing, granulated compounds, formulated from alkaline salts, phosphates, chlorine releasing compounds, and sequestering agents.

3.2 Foam height. The foam height of the compounds in distilled water shall be not greater than 1.5 centimeters (5 second value) when tested as specified in 4.4.

3.3 Composition. The dishwashing compounds shall conform to the requirements shown in table I, when tested as specified in 4.4. The moisture content of the type II, class 2 compounds shall not exceed 2 percent when tested as specified in 4.4.

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TABLE I. Chemical and physical requirements

Classification			Percent phosphate calculated as P_2O_5	Percent carbonate calculated as CO_2	Percent total alkalinity expressed as Na_2O	pH 1% solution	Water softening capacity	Specific gravity at 80°F	Percent avail. chlorine expressed as CL_2	Total solids (%)
Type I	Part A	-	0	-	12.5-14.0	12.5-12.8	2/	Not less than 1.20	Not less than 1.50	17.5-19.0
	Part B	-	16.0-17.5	-	0	7.0-7.2	2/	-	0	28.5-30.0
Type II		Style A	0	Trace	9.0-11.0	12.0-12.5	15 Gr. Hard Water 1/	Not less than 1.15	Not less than 1.75	19.5-22.0
	Class 1	Style B	10.0-14.0	Trace	11.5-12.5	11.9-12.5	15 Gr. Hard Water 1/	Not less than 1.40	0	39.0-41.0
		Style C	0	Trace	18.0-20.0	11.2-12.6	15 Gr. Hard Water 1/	Not less than 1.24	0	27.0-28.5
	Class 2	Style A	37.0-41.0	0	15.0-19.0	11.3-11.6	14 Gr. Hard Water 1/	-	Not less than 1.75	3/
		Style B	21.0-25.0	10.0-14.0	37.0-41.0	11.8-12.5	11 Gr. Hard Water 1/	-	0	3/
		Style D	14.0-18.0	18.0-22.0	31.0-35.0	11.3-12.6	11 Gr. Hard Water 1/	-	0	3/

1/ No precipitate shall form in water up to the hardness specified when tested as specified in 4.5.5.

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2/ A combination of part A and part B compound is necessary for testing, not more than a ratio of 1 : 1 for part B to Class 1 product shall be required to soften the water.

3/ Not applicable.

3.4 Toxicity. The ingredients used in formulating the dishwashing compounds used in liquid dispensing systems shall not be contaminated with toxic amounts of poisonous compounds such as lead, mercury, arsenic, or antimony. The compound shall contain not more than 0.003 percent heavy metals, expressed as lead, when tested as specified in 4.4.

3.5 Corrosion. Steel or aluminum test panels shall show no evidence of pitting, corrosion, etching, or discoloration, and a weight change of more than 1.2 milligrams loss per square centimeter per hour shall not occur when the dishwashing compound is tested as specified in 4.4 and 4.5.

3.6 Soap. The dishwashing compound shall show no evidence of soap when tested as specified in 4.4.

3.7 Particle size. The particle size of the type II, class 2 dishwashing compound shall be generally uniform to minimize segregation of the ingredients. Not less than 95 percent of this type compound shall pass through a size No. 16 sieve and not more than 5 percent shall pass through a size No. 170 sieve when tested as specified in 4.4. Sieves shall conform to RR-S-366.

3.8 Stability in storage. The dishwashing compound shall not lose its original effectiveness or develop a rancid, fishy, or acrid odor when stored for 1 year in a closed shipping container at room temperature (see 6.2.1). Material susceptible to decomposition will be required to contain a preservative or inhibitive to comply with the specified storage life. The type I, part A and part B, and type II, class 1 products shall maintain their chemical stability for 1 year after receipt and show no tendency towards precipitation, sedimentation, or leakage when maintained between temperatures of 45°F and 115°F.

3.9 Labeling. Each container of dishwashing compound or water conditioner component shall be legibly marked (see 5.3.3.1) with black ink, and, labels shall contain the following information, precautions, and instructions for use in accordance with the type, class, and style as applicable, conforming to the Federal Hazardous Substances Act.

3.9.1 Warning label.

3.9.1.1 All type I; part A and part B, and type II; class 1, style A, B, or C items, as applicable, shall be labeled with the following warning information:

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"WARNING! HARMFUL IF SWALLOWED AND CAUSES IRRITATION.
AVOID CONTACT WITH EYES, SKIN AND CLOTHING.
WASH THOROUGHLY AFTER HANDLING."

3.9.1.2 All type II, class 2, style A, B, or D items shall be labeled with the following warning information:

"WARNING! HARMFUL IF INHALED AND CAUSES IRRITATION.
AVOID BREATHING DUST.
KEEP CONTAINER CLOSED.
USE ONLY WITH ADEQUATE VENTILATION
AVOID CONTACT WITH EYES, SKIN AND CLOTHING.
WASH THOROUGHLY AFTER HANDLING."

3.9.2 Caution label.

3.9.2.1 All type I, part A and part B, and type II; class 1, style A, B, or C items, as applicable, shall be labeled with the following caution information:

"CAUTION: ALKALINE CAUSTIC LIQUID. VENTED CONTAINER.
DO NOT DROP.
REMOVE LEAKING PACKAGES.
WASH CAUSTIC ALKALINE LIQUID OFF WITH WATER.
DO NOT LOAD WITH EXPLOSIVES OR NEAR ARTICLES
BEARING YELLOW WARNING LABELS.
DO NOT MIX WITH ACIDS OR OTHER CHEMICALS.
KEEP FROM FREEZING.
STORE ABOVE 45°F."

3.9.2.2 All type II, class 2, style A, B, or D items shall be labeled with the following caution information:

"CAUTION: ALKALINE POWDER. DO NOT MIX WITH ACID OR OTHER
CHEMICALS.
ATTACH ONLY TO WATER INLET SUPPLY AND DISHWASHING
COMPOUND DISPENSER OF SINGLE PRODUCT DISPENSERS.
DO NOT ATTACH TO RINSE INJECTOR PORTION OF DISPENSER."

3.9.3 Special marking. In addition to the marking specified in 3.9.1 and 3.9.2, individual containers of liquid system dishwashing compounds shall contain the following information.

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3.9.3.1 Type I, part A.

"DISHWASHING COMPOUND, LIQUID SYSTEM TYPE I, PART A.
MULTIPLE PRODUCT TYPE.
FOR USE ONLY IN LIQUID TYPE DISPENSERS REQUIRING A TWO-PART SYSTEM
DO NOT USE SEPARATELY FROM WATER CONDITIONER, CONFORMING TO TYPE I PART B OF P-D-1801.
DO NOT ATTACH TO, OR USE IN THE WATER CONDITIONING OR RINSE ADDITIVE PORTION OF THE DISPENSER."

3.9.3.2 Type I, part B.

"WATER CONDITIONING COMPOUND, LIQUID SYSTEM TYPE I, PART B.
MULTIPLE PRODUCT TYPE.
FOR USE ONLY IN LIQUID TYPE DISPENSERS REQUIRING A TWO PART SYSTEM.
DO NOT USE SEPARATELY FROM LIQUID DISHWASHING COMPOUND, CONFORMING TO TYPE I, PART A OF P-D-1801.
THIS IS NOT A CLEANING COMPOUND
DO NOT ATTACH TO, OR USE IN THE DISHWASHING COMPOUND OR RINSE ADDITIVE PORTION OF THE DISPENSER."

3.9.3.3 Type II, class 1, style A, B, or C.

"DISHWASHING COMPOUND LIQUID SYSTEM TYPE II, CLASS 1, STYLE (as applicable).
FOR USE ONLY IN LIQUID TYPE DISPENSERS REQUIRING A SINGLE PRODUCT SYSTEM
DO NOT USE IN COMBINATION WITH ANY OTHER PRODUCT
ATTACH ONLY TO FRESH HOT WATER INLET SIPHON OF DISPENSER
DO NOT ATTACH TO THE RINSE ADDITIVE PORTION OF THE DISPENSER."

3.9.3.4 Type II, class 2, style A, B, or D.

"DISHWASHING COMPOUND, LIQUID SYSTEM TYPE II, CLASS 2, STYLE (as applicable)
FOR USE ONLY IN LIQUID TYPE DISPENSERS REQUIRING A SINGLE PRODUCT SYSTEM HAVING A FRESH HOT WATER MIXER VALVE ON THE DISPENSER.
DO NOT USE IN COMBINATION WITH ANY OTHER PRODUCT
ADD ONLY WATER FROM THE FRESH HOT INLET TO THIS PRODUCT
DO NOT ADD WATER MANUALLY
DO NOT ATTACH TO THE RINSE ADDITIVE PORTION OF THE DISPENSER."

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3.10 Workmanship. The dishwashing compounds covered herein shall be thoroughly mixed, present no evidence of lumping, segregation, separation, layering, crystallizing (liquid products), or thickening. They shall be clean and free of sediment or salt.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.1.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in applicable subparagraphs as follows, and at the inspection levels and acceptable quality levels (AQLs) set forth herein. A lot shall consist of the dishwashing compound manufactured in one batch. The lot shall be expressed in units of filled unit containers, of the same capacity, for examination in 4.3.1.1, and in units of shipping containers for examination for preparation for delivery under 4.3.1.2.

4.3.1.1 Examination of contents of the unit container. The sample unit for this examination shall be the contents of one filled unit container, as applicable. The inspection level shall be S-3 with an AQL of 2.5, expressed in terms of defects per hundred units.

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Examine	Defect
Odor	Rancid, fishy, or acrid.
Appearance	Any evidence of lumping, segregation, separation, layering, crystallizing (liquid), or thickening. Any sediment, salt, or dirt. Not uniform.

4.3.1.2 Examination of preparation for delivery requirements. An examination shall be made to determine that the packaging, packing, and marking complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery, with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 2.5 defects per hundred units. For examination of net contents, one container in the sample shall be examined.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect; illegible; of improper size, location, sequence or method of application
Materials	Any component missing or damaged. Any component not as specified.

Examine	Defect
Workmanship	Inadequate application of components, such as; pallet, shipping containers or sealed case liner not as specified; incomplete closure of container flaps, improper stapling, inadequate taping or loose strapping. Bulged or distorted container.
Contents	Quantity is less than indicated in pail or drum Number of interior packages per shipping container is more or less than required.

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4.4 Testing of the end item. The end item shall be tested for the characteristics indicated in table II for each lot presented for examination for each type, class, and style of material. The sample unit, for purposes of testing, shall be a composite sample selected in accordance with 4.4.1. Unless otherwise specified in table II, two determinations shall be performed on the composite sample. The average of two determinations shall meet the requirements when results are reported numerically. Both determinations shall meet the requirements when the results are reported as pass or fail. The lot size shall be expressed in units of containers of the specified quantity, type, class, and style of dishwashing compound. The sample unit shall be tested with no evidence of failure to meet the applicable requirements. Test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if the composite fails to meet any test requirements specified.

4.4.1 Composite samples.

4.4.1.1 Liquid samples. Sixteen ounces of liquid dishwashing compound shall be drawn from each of five shipping containers, and composited. The test sample unit shall consist of 1/2 gallon of the composite. The sample unit shall be placed in a clean, dry, glass container. The composite sample shall be marked with the identifying lot number and date.

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INSTRUCTIONS FOR TESTING

TABLE II

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations	Results Reported As		Inspect. Level	AQL
	Requirement	Test Method	Individ Unit	Comp Sample	Per Comp Sample	Pass or Fail 2/	Numerically to Nearest		
Phosphate Content, expressed as P_2O_5	Table I	4.5.1		X	Average of 2		0.1 percent		
Percent Carbonates, calculated as CO_2	Table I	4.5.2		X	Average of 2		0.1 percent		
Total Alkalinity, expressed as Na_2O	Table I	4.5.3		X	Average of 2		0.1 percent		
pH	Table I	4.5.4		X	Average of 2		0.1 pH		
Water Softening Capacity	Table I	4.5.5		X	Average of 2				
Specific Gravity	Table I	4.5.6		X	Average of 2	X	0.01 Specific Gravity		
Percent Chlorine, expressed as Cl_2	Table I	4.5.7		X	Average of 2		0.1 percent		
Percent Solids (liquid forms)	Table I	4.5.8		X	Average of 2		0.1 percent		
Moisture Content (solid forms)	3.3	4.5.9		X	Average of 2		0.1 percent		
Toxicity	3.4	4.5.10		X	1	X			
Corrosion	3.5	4.5.11		X	1	X			
Soap test	3.6	4.5.12		X	1	X			
Particle Size	3.7	4.5.13		X	1		0.1 percent		
Stability	3.8		X		Cert of compliance 3/				
Foam Height	3.2	4.5.14			Average of 2		0.1 cm		

1/ Test report shall include all values on which results are based.

2/ If failure is indicated, report either description of failure or numerical point of failure, as applicable.

3/ See 4.1.1

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4.4.1.2 Powdered samples. A sample of not less than 1 pound each shall be taken from not less than four separate shipping containers, selected throughout the lot presented for test. The total samples shall consist of not less than three sample portions taken from each separate shipping container, and shall be placed in a clean, dry container of a suitable size. After thorough mixing, a 1 pound sample shall be drawn for testing purposes. The composite sample shall be placed in an airtight container, sealed, and marked with the identifying lot number and date.

4.5 Tests.

4.5.1 Total phosphates (as P_2O_5). The total phosphate content of the dishwashing compounds shall be determined in accordance with method 1501.0 of FED-STD-536.

4.5.2 Total carbonates. The total carbonate content of the dishwashing compounds shall be determined in accordance with method 1402.1 of FED-STD-536.

4.5.3 Total alkalinity. The total alkalinity of the dishwashing compounds, expressed as Na_2O, shall be determined in accordance with method 304.3 of FED-STD-536.

4.5.4 pH. The pH of a 1.0 percent solution shall be determined as specified in ASTM E-70-68.

4.5.5 Water softening capacity. The water softening capacity of the dishwashing compounds shall be determined as follows:

4.5.5.1 Preparation of stock solutions for test.

4.5.5.1.1 Preparation of stock solution of hard water. Prepare a solution using distilled water which has been freshly boiled and cooled to 20° ± 1°C, containing calcium chloride and magnesium chloride equivalent to 2.85 mg of calcium carbonate and 1.20 mg of magnesium carbonate per ml, respectively. This solution contains 4275 ppm hardness (approximately 250 grains per gallon) as calcium carbonate. It shall be prepared by dilution of more concentrated stock solutions of calcium chloride and magnesium chloride which have been accurately analyzed. The deviation from the specified concentrations shall not exceed 1 part per 100.

4.5.5.1.2 Transfer exactly 15 ml of the stock hard water solution into a 250 ml volumetric flask and add sufficient distilled water to the mark. This will make a stock solution of 15 grain hard water.

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4.5.5.1.3 Transfer exactly 14 ml and 11 ml, respectively, of the stock hard water solution into two more 250 ml volumetric flasks and continue preparation as specified in 4.5.5.1.2. This will give stock hard water solutions of 14 grain and 11 grain hard water, respectively.

4.5.5.2 Procedure.

4.5.5.2.1 Blank solutions. All water softening tests shall be run against blank solutions using distilled water in place of hard water for comparison purposes.

4.5.5.2.2 Type I compounds. An accurately measured amount of type I, part A compound shall be added to 100 ml of the hard water stock solution prepared in 4.5.5.1.2 to make a 0.25 percent solution of dishwashing compound in 15 grain hard water. Not more than a 1 to 1 ratio of class 2 to class 1 compound shall then be added to the solution. Ten ml of this solution shall be placed into a clean, dry, 20 millimeter by 150 millimeter test tube. The solutions shall be heated to 66°C for 5 minutes, removed from the heat and allowed to cool by standing in room temperature for 30 minutes. There shall be no precipitate formation or flock in the bottom of the test tube at the end of the 30 minutes.

4.5.5.2.3 Type 2, class 1, style A, B, or C compounds. Testing shall be performed as specified in 4.5.5.2.2, except that each compound shall be used by itself with no other combination product added.

4.5.5.2.4 Type 2, class 2, style A, B, or D compounds. Testing shall be performed as specified in 4.5.5.2.3, except that 14 or 11 grain hard water solutions prepared as specified in 4.4.1.2 shall be used, as necessary to meet the requirements specified in table I.

4.5.6 Specific gravity. Specific gravity shall be determined in accordance with Method 4183 of FED-STD-141, test 5, Specific Gravity, by means of a Pycnometer. The specific gravity shall be determined at 26.7° $\pm$ .2°C.

4.5.7 Available chlorine. Chlorine content, expressed as Cl_2, shall be determined in accordance with paragraphs 6.174 to 6.175 (arseneous oxide titration) of the Official Methods of Analysis of the Association of Official Analytical Chemists. A 2 gram sample shall be used for type 2, class 2 samples.

4.5.8 Total solids (type I, class 1 and 2 and type II, class 1 compounds only. Weigh 20 to 40 grams of the liquid dishwashing compound and place in a tared evaporating dish, approximately 9 to 10 centimeters in diameter and 2 to 3 centimeters deep, and dry in a constant temperature air oven at 105° $\pm$ 2°C. Constant weight is obtained when successive heatings for a 1 hour period show a loss (or gain) of not more than 0.1 percent.

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4.5.9 Moisture content (type II, class 2 compounds only). The moisture content shall be determined in accordance with method 201.1 of FED-STD-536.

4.5.10 Toxicity.

4.5.10.1 Reagents.

4.5.10.1.1 Standard lead solution. Transfer 0.159 ± 0.001 g of chemically pure (CP) lead nitrate to a 1000 ml volumetric flask. Dilute to mark with distilled water and mix well. Pipette 10 ml of this solution into a 100 ml volumetric flask, add distilled water to the mark, and mix well. This solution contains 0.01 mg lead per ml.

4.5.10.1.2 Sodium hydroxide (app. 2N). Dissolve 80.0 g of CP sodium hydroxide pellets in somewhat less than 1000 ml of distilled water, transfer to a 1000 ml volumetric flask, make up to mark, and mix well.

4.5.10.1.3 Phenolphthalein indicator. Dissolve 1.0 g of phenolphthalein powder in 50 ml of 95 percent ethyl alcohol. Add 50 ml of distilled water and mix.

4.5.10.1.4 Acetic acid (app. 1M). Dilute 57 ml of glacial acetic acid to 1 liter in a volumetric flask.

4.5.10.1.5 Hydrogen sulfide water. Pass hydrogen sulfide into distilled water for 15 minutes.

4.5.10.1.6 Preparation of stock solution of sample. Prepare a solution of the compound by dissolving 50.0 grams in distilled water which has been freshly boiled and cooled to 20° ± 1°C, at 55° to 65°C. Cool to 20° ± 1°C., make up to exactly 1 liter with water and mix thoroughly. This solution shall be allowed to stand for not more than 1 hour to allow any insoluble matter to settle. Approximately 500 ml of the clear supernatant solution shall be decanted into a clean, dry, glass-stoppered bottle.

4.5.10.1.7 Procedure. Pipette 100 ml of unfiltered stock solution of dishwashing compound (see 4.5.10.1.6) into a 250 ml beaker. Slowly add 10 ml of concentrated hydrochloric acid to the solution. Evaporate the solution to complete dryness on a steam bath. Bake the residue in a circulating oven at 105°C. for 2 hours. Add 50 ml of distilled water to the residue and heat briefly on the steam bath to dissolve all soluble salts. Filter the solution quantitatively through a No. 41 Whatman filter paper. (All the insoluble silica need not be transferred to the filter paper.) Wash the beaker twice with 5 ml portions of distilled water and pour through the filter. Finally, wash the filter paper with 5 ml of distilled water. The washings should be collected in the beaker containing the filtrate. Add three drops of phenolphthalein indicator to

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the filtrate and titrate to the end-point with 2N sodium hydroxide. Record the volume of sodium hydroxide used. Add 1M acetic acid dropwise until the color of the indicator just disappears, then add 0.2 ml in excess. Cover the beaker and set the solution aside. Prepare a control solution as follows: Add 15 ml of standard lead solution plus a volume of the 2N sodium hydroxide equal to that used to titrate the sample to a 250 ml beaker. Make up to 100 ml with distilled water and add 10 ml of concentrated hydrochloric acid. Evaporate the solution to dryness and proceed as for the sample solution, except that oven baking time may be reduced to about 30 minutes. Refilter the sample solution if it is not clear, keeping the volume less than 90 ml. Transfer the sample solution and the control solution to separate 100 ml Nessler tubes. Add 10 ml of freshly prepared hydrogen sulfide water to each. Dilute to the mark with distilled water, stopper, and mix well. The color in the sample solution should not exceed that in the control. The color comparisons shall be made in the shortest possible time (5 minutes maximum) after the hydrogen sulfide water has been added.

4.5.11 Corrosion.

4.5.11.1 Procedure for steel. Two strips, each 1 cm by 8 cm, of class 302, class 304, and class 430 corrosion-resistant steels conforming to condition A, finish No. 6 of QQ-S-766 shall be prepared. The strips shall then be degreased by swabbing in a mixture of equal parts alcohol and acetone. Allow the solvent to air evaporate and heat in a circulating air oven at 105° ± 2°C for 30 minutes. Allow the strip to cool to room temperature in a desiccator and upon removal, quickly weigh them to the nearest 0.1 mg. Prepare a solution of 6.5 grams of the dishwashing compound in 350 ml of distilled water. Pour 160 ml of this solution into a 32 mm by 300 mm test tube, completely immerse the sample strips into the solution and maintain the solution and strips at 79.5° ± 2°C for a period of 2 hours. At the end of the 2 hours, rinse the strips thoroughly, first with tap water, and then with acetone. Allow the strips to dry in a circulating air oven at 100° ± 5°C. Allow the strips to cool to room temperature in a desiccator. Remove the strips from the desiccator and quickly reweigh them to the nearest 0.1 mg. Calculate the change in weight of the strips as milligrams per square centimeter per hour of exposed surface. Examine the strips for evidence of corrosion such as pitting, etching or discoloration.

4.5.11.2 Procedure for aluminum. Totally immerse two strips, 8 cm by 2 cm, aluminum-alloy sheet, conforming to temper H14 of QQ-A-250/2 at 82°C. (180°F.) for 5 hours in 250 ml solutions of the compound prepared as specified in 4.4.11.1. Four hundred ml beakers are of convenient size for the solution. Maintain solution level with distilled water. After 5 hours exposure, remove the specimens, rinse with distilled water, and dry. Examine the specimens for discoloration, etching, or formation of a white film on the surface.

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4.5.12 Soap test. Dissolve 2 grams of dishwashing compound in 100 ml of distilled water. Filter off any insoluble matter. Pour 5 ml of the filtrate into a test tube and acidify with glacial acetic acid. The appearance of a fatty acid precipitate or oily droplets indicates the presence of soap.

4.5.13 Particle size.

4.5.13.1 Apparatus. Test shall be made using a single-eccentric-type mechanical shaker which imparts to the sieves a rotary motion and tapping action of uniform speed of 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute.

4.5.13.2 Procedure. Place a 100 g sample on the coarser screen of the nest of appropriate sieves and screen in the shaking apparatus. Continue screening for 5 minute intervals until the weight retained on finer screen is reduced by not more than 0.1 g during successive intervals. Weigh material on the finer screen and calculate percent of material falling within the appropriate range. Tests shall be conducted at a relative humidity not exceeding 50 percent.

4.5.14 Foam height.

4.5.14.1 Equipment.

(a) The stirring motor shall be a standard Hamilton Beach Model #936 mixer, or equal, with the stirring blade attached. The motor, with the stirring blade attached, shall be removed from its original stand and supported on a ring stand by means of a 250 ml beaker clamp.

(b) The test cell shall be a glass cylinder 7.0 cm inside diameter and 18.5 cm high made by sealing one end to provide a flat bottom.

(c) The baffle used shall be a cage constructed as shown in figure 1.

(d) A water bath shall be used to control the temperature ± 0.5°C, in the test cell.

4.5.14.2 Procedure. Two hundred ml of 0.01 percent test solution of the sample dishwashing compound shall be placed in the test cell and allowed to equilibrate in a 66° ± 5°C bath. The stirrer shall be centered between the four baffles and clamped in position so that the liquid level shall be midway between the two blades on the stirrer. The stirring motor shall be turned on and run for 3 minutes at a setting to produce a rated 19,000 rpm, to allow the foam to reach maximum height. The stirrer shall be stopped after running 3 minutes ± 2 seconds, and

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foam height measured in centimeters after 5 seconds. The foam height shall be determined by measuring the total height in centimeters of the combined foam and liquid and the liquid measured prior to agitation. The difference is proportional to the volume of entrapped air and shall be designated as foam height. Distilled or de-ionized water shall be required for the above test.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Dishwashing compound shall be packaged in pails or drums commercially used for the specific product being packaged. The pails and drums used shall neither affect nor be affected by the product being packaged. Unless otherwise specified (see 6.2), the quantity of dishwashing compound per pail or drum package shall be as specified in table III.

TABLE III. Quantity of dishwashing compoundper pail or drum package

Product	Quantity	Package
Type I, part A	19 gallon	Drum
Type I, part B	20 gallon	Drum
Type II, class 1, styles A, B, and C	6 gallon	Pail
	15 gallon	Drum
	30 gallon	Drum
Type II, class 2, styles A, B, and D	50 pound	Pail

5.1.2 Level B (Civil Agencies). Dishwashing compound shall be packaged in accordance with the applicable requirements specified in 5.1.1 (see 6.2).

5.1.3 Level C. Dishwashing compound shall be packaged to afford adequate protection against physical damage and deterioration during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

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5.2.1.1 When specified (see 6.2), dishwashing compound of one description only, shall be packed on pallets in accordance with load type XIII of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L.

5.2.1.2 When specified (see 6.2), dishwashing compound, packaged as specified in 5.1, shall be individually packed in shipping containers as hereinafter specified.

5.2.1.2.1 Pails. Each pail of dishwashing compound shall be packed upright in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each fiberboard container shall be fitted with a taped box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. When applicable, the pail shall be immobilized within the container by means of die-cut or built-up fiberboard pads fabricated of the same material as the shipping container. Each shipping container shall be closed in accordance with method III and reinforced as specified in the appendix of the container specification.

5.2.1.2.2 Drums. Each drum of dishwashing compound shall be packed upright in a snug-fitting wood shipping container conforming to class 2, style 1 of PPP-B-585 or overseas type, style A, grade A or B of PPP-B-601. When applicable, the drum shall be immobilized within the container by means of blocking and bracing. Each wood container containing a fiber drum shall be provided with a sealed case liner conforming to type I or II, grade C of MIL-L-10547. Each shipping container shall be closed and reinforced in accordance with the appendix of the applicable container specification.

5.2.2 Level B.

5.2.2.1 When specified (see 6.2), dishwashing compound of one description only, packaged as specified in 5.1, shall be packed on pallets as specified in 5.2.1.1.

5.2.2.2 When specified (see 6.2), dishwashing compound, packaged as specified in 5.1, shall be individually packed in shipping containers as hereinafter specified.

5.2.2.2.1 Pails. Each pail of dishwashing compound shall be packed upright in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each fiberboard container shall be fitted with a taped box liner

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conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. When applicable, the pail shall be immobilized within the container by means of die-cut or built-up fiberboard pads fabricated of the same material as the shipping container. Each shipping conatiner shall be closed in accordance with method II as specified in the appendix of the container specification.

5.2.2.2.1.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.2.2.2 Drums. Each drum of dishwashing compound shall be packed upright in a snug-fitting wood shipping container conforming to class 1, style 1 of PPP-B-585 or domestic type, grade B of PPP-B-601. When applicable, the drum shall be immobilized within the container by means of blocking and bracing. Each shipping container shall be closed and reinforced in accordance with the appendix of the applicable container specification.

5.2.3 Level C. Dishwashing compound, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Civil Agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Special marking.

5.3.3.1 All interior packages and shipping containers. All interior packages and shipping containers shall be marked with the information in 3.9.

5.3.3.2 All shipping containers, except pallets. The top panel of each shipping container, except pallets, shall be marked with the word "TOP". In addition, two side panels of each shipping container, except pallets, shall be affixed with the applicable special marking requirements for arrows as specified in MIL-STD-129.

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6. NOTES

6.1 Intended use. The dishwashing compounds covered by this specification are intended for use only in liquid type dispensing systems. They are not to be used in powder dispenser reservoirs or for hand feeding operations. Care should be observed in selecting the correct compound needed for the particular dispenser present on the dishwasher. Dishwashing compounds are not directly interchangeable with all manufacturer's dispensing equipment or with all equipment from the same manufacturer.

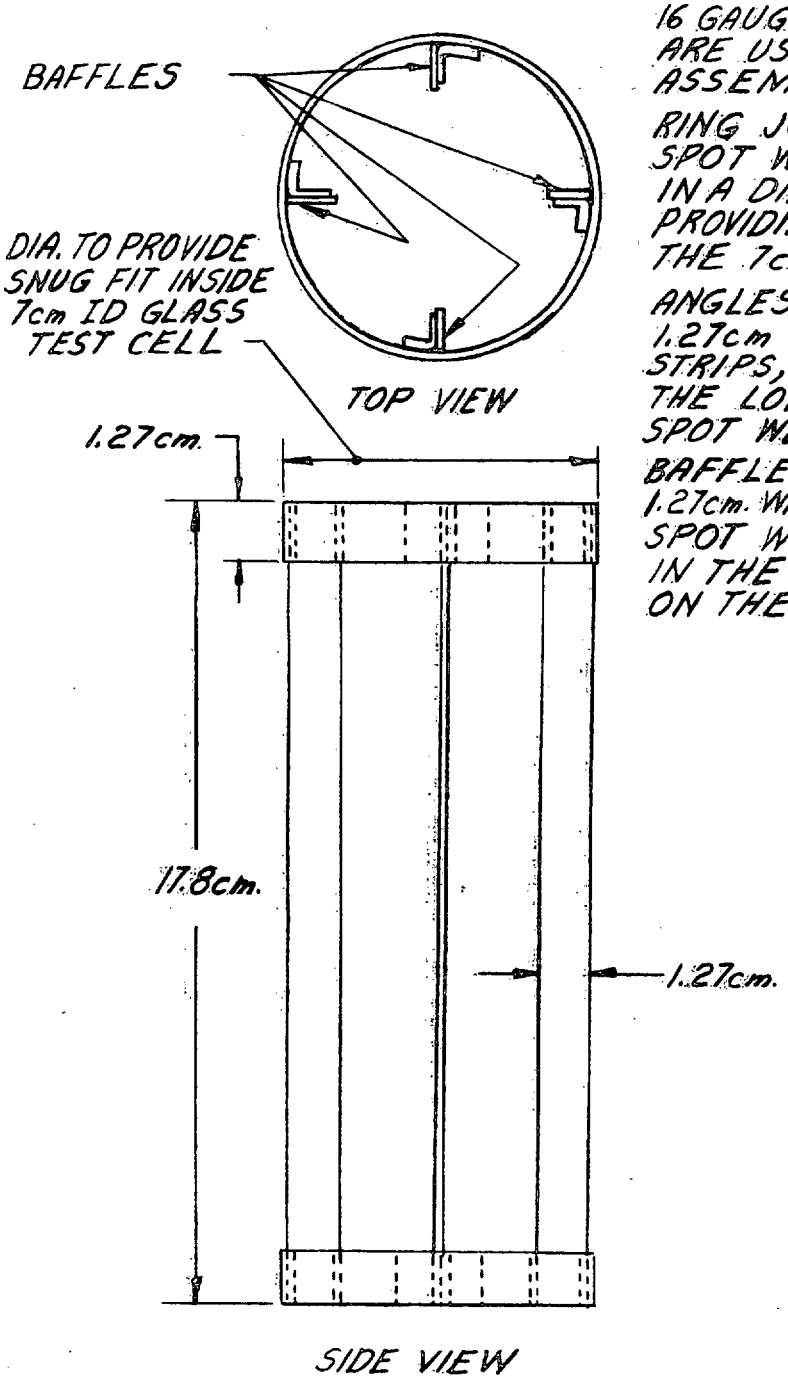
6.2 Ordering data. Purchasers should exercise any desired options permitted herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, part, class, style and quantity of dishwashing compound required (see 1.2).
(c) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When level B packaging is required for civil agencies (see 5.1.2).
(e) When interior packages are to be packed on pallets (see 5.2.1.1 and 5.2.2.1).
(f) When interior packages are desired to be individually packed in shipping containers (see 5.2.1.2 and 5.2.2.2).
(g) When weather-resistant grade fiberboard shipping containers are desired for the level B individual packing of pails (see 5.2.2.2.1.1).

6.2.1 The following statement should be included in the purchase contract: "If latent defects should be discovered for a period of 13 months after material has been accepted, the Government reserves the right to reject the delivery, require replacement, or cancel the contract and surcharge the contractor for any expense incurred by the cancellation of the contract and in securing satisfactory material".

6.3 Basis of purchase. Type I, part A and part B, and type II, class 1, style A, B, or C should be purchased by net volume. Type II, class 2, style A, B, or D should be purchased by net weight.

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16 GAUGE STAINLESS STEEL STRIPS ARE USED THROUGHOUT THE ASSEMBLY.

RING JOINTS ARE LAPPED, SPOT WELDED AND FEATHERED IN A DIAMETER CAPABLE OF PROVIDING A SNUG FIT INSIDE THE 7cm ID GLASS TUBING.

ANGLES ARE FORMED BY USING 1.27cm WIDE BY 2.54cm LONG STRIPS, BENT IN THE CENTER OF THE LONG DIRECTION AND SPOT WELDED TO THE FORMED RINGS.

BAFFLES ARE FORMED USING 1.27cm. WIDE BY 17.8cm LG STRIPS, SPOT WELDED TO THE ANGLES IN THE POSITIONS INDICATED ON THE DIAGRAM.

FIG 1 BAFFLE CAGE FOR FOAM HEIGHT DETERMINATION

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Custodians:

Army - GL
Air Force - 84

Review activity:

Army - MD

User activities:

Navy - MS, SA

Preparing activity:

Army - GL

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
HEW-FDA
HEW-NIH

Project No. 7930-0278

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION P-D-1801

Dishwashing Compound, Liquid System (For Commercial Dishwashing Machines)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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